

ENGE702 Engineering Mathematics III

Tutorial 3

1. Find the general solution to the following differential equations:

(a) $y'' - 3y' + 2y = 0$ $y = C_1 e^x + C_2 e^{2x}$ (c) $y'' - 2y' + y = 0$ $y = C_1 e^x + C_2 x e^x$
 (b) $2y'' + 8y' + 6y = 0$ $y = C_1 e^{-3x} + C_2 e^{-x}$ (d) $y'' + 3y' + 5y = 0$ $y = e^{-\frac{3}{2}x} \left(A \sin\left(\frac{\sqrt{11}}{2}x\right) + B \cos\left(\frac{\sqrt{11}}{2}x\right) \right)$

2. Solve the following initial value problems:

(a) $y'' - 7y' + 10y = 0$, $y(0) = 1$, $y'(0) = 2$. $y = e^{2x}$
 (b) $y'' + 6y' + 9y = 0$, $y(0) = 0$, $y'(0) = -1$. $y = -x e^{-3x}$
 (c) $y'' + 5y' + 7y = 0$, $y(0) = 1$, $y'(0) = 0$. $y = e^{-\frac{5}{2}x} \left(\frac{5\sqrt{3}}{3} \sin\left(\frac{\sqrt{3}}{2}x\right) + \cos\left(\frac{\sqrt{3}}{2}x\right) \right)$

3. Suppose we have an unforced mass-spring-damper system with damping coefficient $c > 0$. What state will the system reach in the long-term? Explain why both mathematically and physically. *It will return to equilibrium because: $e^{-\alpha t}$ number $\rightarrow$ energy removed from damper*

4. Suppose we have an unforced mass-spring-damper system with mass $m = 1$, damping factor $c = 1$ and spring constant $k = 0$ (really this is just a mass-damper system). Suppose the initial displacement is $y(0) = 1$ and the initial velocity is $y'(0) = \alpha$. Describe how various values of α affect the solution, both mathematically and physically (particularly when $\alpha = 0$). $y = C_1 e^{-x} + C_2$ where $C_1 = -\alpha$, $C_2 = 1 + \alpha$. $\alpha = 0$: no motion, $\alpha > 0$: decays, $\alpha < 0$: no equilibrium

5. Suppose we have an unforced mass-spring-damper system with mass $m = 4$, and spring constant $k = 7$. Find a value for the damping coefficient c so the system is critically damped, and then find the general solution of the system. Then find an initial condition that allows the mass to 'overshoot' the equilibrium position. *Critical damping: $c = 8\sqrt{\frac{7}{4}}$, $y = C_1 e^{-\frac{\sqrt{7}}{2}x} + C_2 x e^{-\frac{\sqrt{7}}{2}x}$*

6. The angle a pendulum makes with the vertical can be modelled as $\theta'' + \frac{g}{l} \sin(\theta) = 0$, *eg let $y(0) = 1$, need $C_2 < -1$, then any $y'(0) < -\sqrt{\frac{g}{l}}$ will do.*

where l is the length of the pendulum, and g is the gravitational constant. Find a way to linearize the model, and find the general solution (*hint: for small values of θ , we have $\sin(\theta) \approx \theta$*). *for small θ , $\theta \approx \sin(\theta)$, so linearise $\theta'' + \frac{g}{l}\theta = 0$*

7. Finished all these? Work through problem sets 1.3 and 1.5 in the textbook

Then $\theta = A \cos(\sqrt{\frac{g}{l}}t) + B \sin(\sqrt{\frac{g}{l}}t)$

Computer Part

Even though tutorials are not run in computer labs, during weeks 1 - 6 of the semester, you should work through the section 'Solving Ordinary Differential Equations With MATLAB' (under 'Computational Mathematics') in the self paced courses on the math-works website, as 'homework'.

If you have never used matlab before, it would be a good idea to work through the section 'MATLAB Onramp' first. Details of how to access these are as follows:

AUT has a licence with the matlab creators that allows you to access an array of online training material, and also download the latest version of matlab to your personal computer (if you wish - you don't need to do this for this course if you don't want to, you can access matlab on the computers at AUT).

To access these resources you will need to create an account with mathworks (the company that produces matlab).

You can do this by visiting [mathworks.com](https://www.mathworks.com) (or google 'matlab'). You can create an account by clicking the 'login' icon on the top right. **It is important you use your AUT email address to create the account, or you will not be able to access the material.**

From your account page, click 'Self-Paced Courses'.